

# Half-One Metrics and the Vertices of the Metric Body: Density, Enumeration, and Angle

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## Abstract

A half-one metric on  $n$  points is a bounded-by-1 pseudometric taking only the values  $\frac{1}{2}$  and 1. Every such vector is automatically a metric, and by the criterion of the companion paper it is an extreme point of the metric body exactly when every component of its edge graph contains an odd cycle. We ask how much of the metric body these metrics account for, and find that the question has three inequivalent readings with three different kinds of answer.

Among half-one metrics themselves the answer is a theorem: the proportion that are extreme tends to 1, exponentially fast. Among all vertices of the metric body it is data: we determine  $|\text{ex}(\bar{\mathcal{M}}_n)|$  exactly for  $n \leq 6$ , obtaining 19, 259 and 27263 with denominator profiles, and the proportion of vertices that are half-one is not monotone. By exterior solid angle it is a conjecture, supported here by an experiment carried far enough to exhibit a crossover near  $n = 18$  that shorter ranges miss.

Along the way we correct the  $q$ -level feasibility density to  $P_q = \frac{1}{2} + \frac{1}{2q^2}$ , record the denominators that arise in upper-half decompositions, and pose a bound on them that is uniform in  $n$ .

## 1 Introduction

Let  $\bar{\mathcal{M}}_n$  denote the metric body: the polytope of bounded-by-1 pseudometrics on  $\mathbf{n} = \{1, \dots, n\}$ , embedded in  $\mathbb{R}^m$  with  $m = \binom{n}{2}$ . Its extreme points are the irreducible building blocks of all metrics, and identifying them is the central difficulty of the subject [1, 2, 4, 8].

Among them the simplest after the partition metrics are the *half-one metrics*.

**Definition 1.1.** A *half-one metric* on  $n$  points is a  $d \in \bar{\mathcal{M}}_n$  with  $d \in \{\frac{1}{2}, 1\}^{\binom{n}{2}}$ . Write  $\mathcal{H}_n$  for this set.

Every vector in  $\{\frac{1}{2}, 1\}^{\binom{n}{2}}$  is a metric, since  $d_{ij} + d_{jk} \geq \frac{1}{2} + \frac{1}{2} = 1 \geq d_{ik}$  always, so  $|\mathcal{H}_n| = 2^{\binom{n}{2}}$  and half-one metrics enjoy an automatic feasibility no other denominator class has. The companion paper [5] determines which are extreme:  $d \in \mathcal{H}_n$  is extreme in  $\bar{\mathcal{M}}_n$  if and only if every component of its edge graph  $\Gamma_d$  contains an odd cycle.

This paper asks how much of  $\bar{\mathcal{M}}_n$  the half-one metrics account for. The interest is that the question has three readings, and they do not answer alike.

**(i) Among half-one metrics.** Almost all of them are extreme. This is Theorem 3.1: the proportion tends to 1, with an explicit exponential bound on the deficit. The proof reads a random half-one metric as an Erdős–Rényi graph  $G(n, \frac{1}{2})$  and shows that three events of high probability—an induced claw somewhere, a local mending condition at every vertex, and connectivity—together force the edge graph to be connected and to carry an odd cycle.

(ii) **Among all vertices of  $\bar{\mathcal{M}}_n$ .** Here we have data rather than a theorem. Section 4 determines  $|\text{ex}(\bar{\mathcal{M}}_n)|$  for  $n \leq 6$  by exhaustive search over rational grids with exact rank certification:

$$|\text{ex}(\bar{\mathcal{M}}_4)| = 19, \quad |\text{ex}(\bar{\mathcal{M}}_5)| = 259, \quad |\text{ex}(\bar{\mathcal{M}}_6)| = 27263,$$

with the denominators occurring at  $n = 6$  distributed as 203, 15090, 7290, 4680 over 1, 2, 3, 4. The extreme half-one metrics number 5, 168, 12326, so their share of all vertices runs 26%, 65%, 45%: not monotone, and the trend beyond  $n = 6$  is unknown.

(iii) **By exterior solid angle.** The probability that a given vertex maximizes a uniformly random linear objective is proportional to the volume of its dual cone met with the unit sphere. Section 7 reports the experiment over  $5 \leq n \leq 40$ : partition metrics dominate through  $n \approx 12$ , the half-one share crosses theirs near  $n = 18$ , and reaches 1 by  $n = 40$ . The Gauss–Bonnet conjecture (Conjecture 7.1) asserts that half-ones carry asymptotically all of the exterior angle. A range stopping at  $n = 10$  sees none of this, which is worth recording because such a range has been read as evidence against the trend.

That (i) is a theorem, (iii) a conjecture, and (ii) neither—and that (ii)’s numbers do not obviously support (iii)—is the shape of the subject as we find it.

Sections 5 and 6 treat the complement of the question: what the vertices that are *not* half-one look like, and why they are scarce. Section 5 records the denominators arising in decompositions of the upper half of  $\bar{\mathcal{M}}_n$  and conjectures a bound on them uniform in  $n$ ; Section 6 computes the density of  $q$ -level points that are metrics, corrects that density to  $P_q = \frac{1}{2} + \frac{1}{2q^2}$ , and derives the vanishing of  $\text{Vol}(\bar{\mathcal{M}}_n)$ .

Throughout, computations are exact unless stated otherwise; the paper, the formalization, and every script behind the tables are archived on Zenodo at [doi:10.5281/zenodo.21953234](https://doi.org/10.5281/zenodo.21953234), with the development repository at [github.com/DavidVFeldman/half-one-metrics](https://github.com/DavidVFeldman/half-one-metrics).

Much of the paper is formally verified. A Lean 4 development accompanying this article (`HalfOne`, in the archive above) proves, over  $\mathbb{Q}$  and from a perturbation-form definition of extremality: the perturbation lemma (Lemma 2.1), Proposition 2.2, Theorem 3.10 and Corollary 3.11, the twin characterization and the Bonferroni bound of Theorem 3.4 as an exact cardinality inequality, the counting identity behind Proposition 6.1, Proposition 4.3, and, for Theorem 3.1, both its deterministic core (connectivity, mending and a claw force the criterion) and the three counting bounds of its proof, assembled into the finite- $n$  deficit inequality. The census counts of  $|\text{ex}(\mathcal{H}_n)|$  are theorems of the development for  $n \leq 7$ : a checker is proved equivalent to the bipartite-component criterion and then executed inside the proof assistant, by kernel reduction for  $n \leq 5$  and by the compiled evaluator (Lean’s `native_decide`, which enlarges the trusted base by the compiler) for  $n = 6, 7$ . The criterion itself (Theorem 2.4) is the companion paper’s theorem and is quoted, not reproved: the formal census counts graphs satisfying the combinatorial predicate. One observation surfaced by the formalization is worth recording: in Proposition 4.3, positive definiteness is not needed for the pullback direction—the pullback of any vertex along any quotient map is a vertex, the zero coordinates pinning every perturbation to a pullback; positive definiteness is what makes the descent canonical.

## 2 Preliminaries

We follow [5]. For  $d \in \bar{\mathcal{M}}_n$  write  $\mathcal{U}_d = \{E : d_E = 1\}$  for the unital edges and  $\mathcal{N}_d$  for the rest. A triangle is *degenerate* when one of its sides equals the sum of the other two. The *edge graph*  $\Gamma_d$  has node set  $\mathcal{N}_d$ , with  $E$  and  $E'$  adjacent when they are the two short sides of a degenerate triangle.

A point of  $\bar{\mathcal{M}}_n$  is extreme exactly when it is the unique metric satisfying its tight constraints, so extremality is detected by *perturbations*: symmetric  $\varepsilon$  vanishing on the diagonal with  $d \pm \varepsilon \in \bar{\mathcal{M}}_n$ . Two facts about them are used constantly, and we record their one-line proofs rather than import them.

**Lemma 2.1.** *Let  $\varepsilon$  be a perturbation at  $d \in \bar{\mathcal{M}}_n$ . Then*

(i)  $|\varepsilon_E| \leq \min\{d_E, 1 - d_E\}$  for every  $E$ ; in particular  $\varepsilon$  vanishes wherever  $d$  is 0 or 1.

(ii) If the triangle on  $i, j, k$  is degenerate at  $d$ , say  $d_{ik} = d_{ij} + d_{jk}$ , then  $\varepsilon_{ik} = \varepsilon_{ij} + \varepsilon_{jk}$ .

*Proof.* (i)  $d \pm \varepsilon \in \bar{\mathcal{M}}_n$  forces  $0 \leq d_E \pm \varepsilon_E \leq 1$ . (ii) The triangle inequality for  $d + \varepsilon$  gives  $\varepsilon_{ik} \leq \varepsilon_{ij} + \varepsilon_{jk}$  and for  $d - \varepsilon$  the reverse.  $\square$

For a half-one metric the situation is especially simple, because  $\frac{1}{2} + \frac{1}{2} = 1$  is the only degeneracy available.

**Proposition 2.2.** *A strictly positive  $d \in \bar{\mathcal{M}}_n$  has at most one degenerate ordered triangle on any given triple of points.*

*Proof.* Suppose  $d_{ik} = d_{ij} + d_{jk}$  and  $d_{ij} = d_{ik} + d_{kj}$ . Adding gives  $d_{jk} + d_{kj} = 0$ , so  $d_{jk} = 0$ .  $\square$

**Proposition 2.3.** *Let  $d \in \mathcal{H}_n$  and let  $G_d$  be the graph on  $\mathbf{n}$  whose edges are the half-length pairs. Then  $\{i, j\}$  and  $\{i, k\}$  are adjacent in  $\Gamma_d$  if and only if both are edges of  $G_d$  and  $\{j, k\}$  is not.*

*Proof.* A degenerate triangle of a half-one metric has  $\frac{1}{2} + \frac{1}{2} = 1$ , so its two short sides are half-length and its long side unital. Conversely two half-length sides meeting at  $i$  with unital third side form such a triangle.  $\square$

So  $\Gamma_d$  is obtained from the line graph of  $G_d$  by deleting the adjacencies that come from triangles of  $G_d$ . We quote the criterion.

**Theorem 2.4** ([5]).  *$d \in \mathcal{H}_n$  is extreme in  $\bar{\mathcal{M}}_n$  if and only if every component of  $\Gamma_d$  contains an odd cycle. Equivalently, no component of  $\Gamma_d$  is bipartite; isolated nodes count as components.*

### 3 Almost every half-one metric is extreme

Identifying  $d \in \mathcal{H}_n$  with its half-length graph  $G_d$  makes the uniform distribution on  $\mathcal{H}_n$  the Erdős–Rényi distribution  $G(n, \frac{1}{2})$ .

**Theorem 3.1.** *Let  $d$  be a uniformly random half-one metric on  $n$  points. Then*

$$\Pr[d \notin \text{ex}(\mathcal{H}_n)] \leq \left(\frac{15}{16}\right)^{\lfloor n/4 \rfloor} + \frac{n^3}{2} \left(\frac{7}{8}\right)^{n-3} + \sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n}{k} 2^{-k(n-k)},$$

and in particular

$$\lim_{n \rightarrow \infty} \frac{|\text{ex}(\mathcal{H}_n)|}{2^{\binom{n}{2}}} = 1.$$

*Proof.* Write  $G = G_d$ . By Theorem 2.4 it suffices to exhibit events forcing  $\Gamma_d$  to be connected and to contain an odd cycle.

*An odd cycle.* Suppose  $G$  contains an induced claw: a vertex  $i$  with neighbours  $j, k, l$  pairwise non-adjacent. By Proposition 2.3 the three nodes  $\{i, j\}, \{i, k\}, \{i, l\}$  are pairwise adjacent in  $\Gamma_d$ , a triangle. To bound the probability that no induced claw exists, partition  $\mathbf{n}$  into  $\lfloor n/4 \rfloor$  disjoint 4-sets. A fixed 4-set spans an induced claw with probability  $4 \cdot 2^{-6} = \frac{1}{16}$ : six pairs are prescribed, and the four choices of centre give disjoint events. Distinct 4-sets involve disjoint pairs, hence independent events, and the first term follows.

*Local mending.* Call  $i$  *mended* when for all  $j, k \in N_G(i)$  the nodes  $\{i, j\}$  and  $\{i, k\}$  lie in a common component of  $\Gamma_d$ . If  $\{j, k\} \notin G$  they are adjacent outright. If  $\{j, k\} \in G$ , any  $l \notin \{i, j, k\}$  with

$$\{i, l\} \in G, \quad \{j, l\} \notin G, \quad \{k, l\} \notin G$$

supplies the path  $\{i, j\} \sim \{i, l\} \sim \{i, k\}$ . For a fixed triple these three conditions hold for a given  $l$  with probability  $\frac{1}{8}$ , independently over the  $n - 3$  candidates and of the pairs inside  $\{i, j, k\}$ . There are at most  $n \binom{n-1}{2} \leq n^3/2$  triples, giving the second term.

*Connectivity.* The third term is the standard union bound on  $\Pr[G \text{ disconnected}]$  over vertex subsets spanning no crossing pair.

Suppose all three events hold. Any two edges of the connected graph  $G$  are joined by a walk in which consecutive edges share a vertex, and consecutive edges at a mended vertex lie in a common component of  $\Gamma_d$ ; hence  $\Gamma_d$  is connected. It contains the triangle furnished by the claw, so by Theorem 2.4  $d$  is extreme.  $\square$

*Remark 3.2.* The bound is far from sharp; it is a limit statement whose constants bite only for large  $n$ . Exhaustive enumeration gives

| $n$ | $2^{\binom{n}{2}}$ | $ \text{ex}(\mathcal{H}_n) $ | proportion |
|-----|--------------------|------------------------------|------------|
| 4   | 64                 | 5                            | 0.0781     |
| 5   | 1024               | 168                          | 0.1641     |
| 6   | 32768              | 12326                        | 0.3762     |
| 7   | 2097152            | 1309868                      | 0.6246     |
| 8   | 268435456          | 209717144                    | 0.7813     |

and sampling puts the proportion above 0.99 already at  $n = 15$ , where it reads 0.9937 against 0.9889 at  $n = 14$ . How far from sharp the bound is can be said exactly: the right-hand side of Theorem 3.1 exceeds 1, and so says nothing, for every  $n \leq 103$ , the term  $\frac{n^3}{2}(\frac{7}{8})^{n-3}$  dominating until about  $n = 150$  and  $(\frac{15}{16})^{\lfloor n/4 \rfloor}$  thereafter. The theorem is a limit statement, and the range in which anything is computable lies entirely inside the range where its explicit bound is vacuous. The deficit is examined next, and its leading term identified.

The commonest single obstruction to extremality is an isolated node of  $\Gamma_d$ , and it has a graph-theoretic name.

**Proposition 3.3.** *For  $d \in \mathcal{H}_n$ , an edge  $\{i, j\}$  of  $G_d$  is an isolated node of  $\Gamma_d$  if and only if  $i$  and  $j$  are true twins in  $G_d$ : adjacent, with identical neighbourhoods elsewhere.*

*Proof.* By Proposition 2.3 the node  $\{i, j\}$  acquires a neighbour at  $k$  exactly when one of  $\{i, k\}, \{j, k\}$  lies in  $G_d$  and the other does not.  $\square$

An isolated node is a bipartite component, so a true twin pair certifies non-extremality, and counting twin pairs bounds the deficit from below.

**Theorem 3.4.**

$$1 - \frac{|\text{ex}(\mathcal{H}_n)|}{2^{\binom{n}{2}}} \geq \binom{n}{2} 2^{-(n-1)} \left(1 - O(n^2 2^{-n})\right).$$

*Proof.* For a pair  $\{i, j\}$  let  $A_{ij}$  be the event that  $i, j$  are true twins in  $G \sim G(n, \frac{1}{2})$ : the pair  $\{i, j\}$  is present, and for each of the  $n - 2$  remaining vertices  $k$  the pairs  $\{i, k\}$  and  $\{j, k\}$  agree. These are  $n - 1$  independent fair events, so  $\Pr[A_{ij}] = 2^{-(n-1)}$ .

For the second Bonferroni term, two cases. If the pairs share a vertex, say  $A_{ij}$  and  $A_{ik}$ : the twin condition of  $\{i, j\}$  applied at  $k$  forces  $\{j, k\} \in G$ , so all three pairs among  $i, j, k$  are present, and for each of the  $n - 3$  outside vertices  $x$  the three pairs  $\{i, x\}, \{j, x\}, \{k, x\}$  agree; hence  $\Pr[A_{ij} \cap A_{ik}] = 2^{-3} \cdot 2^{-2(n-3)} = 2^{-(2n-3)}$ . If the pairs are disjoint,  $A_{ij}$  and  $A_{kl}$ : the pairs  $\{i, j\}$  and  $\{k, l\}$  are present, each of the  $n - 4$  outside vertices contributes two agreements, and on the four pairs between  $\{i, j\}$  and  $\{k, l\}$  the two twin conditions force all four to agree, leaving one free bit of four; hence  $\Pr[A_{ij} \cap A_{kl}] = 2^{-2} \cdot 2^{-2(n-4)} \cdot 2^{-3} = 2^{-(2n-3)}$ . Both formulas were confirmed by direct enumeration at  $n = 5$ .

There are fewer than  $n^4/8$  unordered pairs of distinct pairs, so by Bonferroni

$$\Pr\left[\bigcup A_{ij}\right] \geq \binom{n}{2} 2^{-(n-1)} - \frac{n^4}{8} 2^{-(2n-3)} = \binom{n}{2} 2^{-(n-1)} \left(1 - O(n^2 2^{-n})\right). \quad \square$$

Exhaustive classification of every non-extreme half-one metric through  $n = 8$  shows the twin obstruction becoming dominant:

| $n$ | non-extreme | with a twin pair | other   | twin share |
|-----|-------------|------------------|---------|------------|
| 4   | 59          | 32               | 27      | 0.54       |
| 5   | 856         | 436              | 420     | 0.51       |
| 6   | 20442       | 11292            | 9150    | 0.55       |
| 7   | 787284      | 545784           | 241500  | 0.69       |
| 8   | 58718312    | 49826744         | 8891568 | 0.85       |

Among the remainder the smallest bipartite component most often has two nodes, an event of probability  $O(n^3 4^{-n})$  per configuration, and every mode observed decays at least this fast against the twin rate. Accordingly:

*Conjecture 3.5.*  $1 - \frac{|\text{ex}(\mathcal{H}_n)|}{2^{\binom{n}{2}}} \sim \binom{n}{2} 2^{-(n-1)}$  as  $n \rightarrow \infty$ .

Writing  $D_n$  for the deficit and  $\lambda_n = \binom{n}{2} 2^{-(n-1)}$  for the twin rate, the conjecture asserts  $D_n/\lambda_n \rightarrow 1$ . That ratio is not monotone: it rises, crosses 1 near  $n = 8$ , and returns to 1 from below.

| $n$ | $D_n$    | $\lambda_n$ | $D_n/\lambda_n$ | warrant |
|-----|----------|-------------|-----------------|---------|
| 4   | 0.921875 | 0.750000    | 1.2292          | exact   |
| 5   | 0.835938 | 0.625000    | 1.3375          | exact   |
| 6   | 0.623840 | 0.468750    | 1.3309          | exact   |
| 7   | 0.375406 | 0.328125    | 1.1441          | exact   |
| 8   | 0.218743 | 0.218750    | 1.0000          | exact   |
| 9   | 0.134295 | 0.140625    | 0.9550          | sampled |
| 10  | 0.084368 | 0.087891    | 0.9599          | sampled |
| 11  | 0.052074 | 0.053711    | 0.9695          | sampled |
| 12  | 0.031611 | 0.032227    | 0.9809          | sampled |
| 13  | 0.018767 | 0.019043    | 0.9855          | sampled |
| 14  | 0.011060 | 0.011108    | 0.9956          | sampled |
| 15  | 0.006339 | 0.006409    | 0.9891          | sampled |
| 16  | 0.003653 | 0.003662    | 0.9974          | sampled |

The sampled rows use  $3 \times 10^6$  draws each, giving a standard error on the ratio that runs from 0.0014 at  $n = 9$  to 0.0095 at  $n = 16$ ; the estimator was checked against the exact values at  $n = 6, 7, 8$ . Every sampled row lies above the floor  $1 - \frac{n^4}{8}2^{-(2n-3)}/\lambda_n$  guaranteed by Theorem 3.4, which is the one consistency check the data owe the proof. The agreement at  $n = 8$  is a crossing and not an identity; the two columns are equal there to five decimal places by coincidence.

*Remark 3.6.* The deficit was previously observed to fit  $c(0.6)^n$  over the computable range, a rate matching no candidate obstruction. The mystery is an artifact of fitting a pure exponential to a sequence with a polynomial factor. Fit  $\lambda_n = \binom{n}{2}2^{-(n-1)}$  itself: the geometric mean of successive ratios over  $4 \leq n \leq 17$  is 0.636, and over  $8 \leq n \leq 17$  it is 0.596; log-linear least squares on the same ranges returns 0.627 and 0.594. Every one of these is an apparent base near 0.6 for a sequence whose actual base is  $\frac{1}{2}$ . The observed 0.6 is the twin law in disguise.

*Remark 3.7.* The proof can be recast, with a better rate, by observing that the nodes  $\{\{i, x\} : x \in N_G(i)\}$  induce in  $\Gamma_d$  exactly the *complement* of  $G[N_G(i)]$ : two spokes at  $i$  are adjacent precisely when the connecting pair is unital. Conditioned on  $N_G(i)$  that complement is again  $G(\deg i, \frac{1}{2})$ , and  $\deg i \geq n/3$  holds for all  $i$  with probability  $1 - e^{-\Omega(n)}$ . An induced claw at  $i$  is a triangle in that complement.

Theorem 3.1 settles [6, Conjecture 4.1.4], whose printed statement asserts  $|\mathcal{H}_n| \sim 2^{\binom{n}{2}}$  and is to be read with  $\text{ex}(\mathcal{H}_n)$  in place of  $\mathcal{H}_n$ : the former is an identity, every element of  $\{\frac{1}{2}, 1\}^{\binom{n}{2}}$  being a metric. It also subsumes the earlier counting bounds. Cayley's formula and Proposition 3.8 below give  $|\text{ex}(\mathcal{H}_n)| > n^{n-2} - \frac{n!}{2}$ , of order  $2^{n \log n}$ ; the truth is  $(1 - o(1))2^{\binom{n}{2}}$ .

**Proposition 3.8.** *If  $G_d$  is a spanning tree that is not a path, then  $d$  is extreme. More generally, if  $G_d$  is triangle-free and every component has a vertex of degree at least 3, then  $d$  is extreme.*

*Proof.* A triangle-free  $G_d$  has  $\Gamma_d$  equal to the line graph of  $G_d$  by Proposition 2.3, and the line graph of a connected graph is connected, so the components of  $\Gamma_d$  correspond to those of  $G_d$ . A vertex of degree 3 contributes a triangle to the line graph. Apply Theorem 2.4.  $\square$

*Remark 3.9.* The resulting inequality is strict at every  $n$ , since the all-unital metric  $d \equiv 1$  is extreme and is counted by no spanning tree; at  $n = 4$  it is the only extreme half-one beyond the four stars, so  $|\text{ex}(\mathcal{H}_4)| = 5$  against  $n^{n-2} - \frac{n!}{2} = 4$ . Comparison with the Bell numbers  $B_n$ , which count the partition metrics, gives  $|\text{ex}(\mathcal{H}_n)| > B_n$  for all  $n \geq 5$  by the bound of Berend and Tassa [3]; at  $n = 4$  the inequality fails,  $5 < 15$ .

The half-one metrics are not merely a convenient family; two results locate them exactly. The first is a corollary of the cube identity of [5] rather than a new fact about  $\bar{\mathcal{M}}_n$ , but it is not stated there and is used repeatedly below.

**Theorem 3.10.** *Every extreme point of  $\bar{\mathcal{M}}_n$  all of whose distances are at least  $\frac{1}{2}$  is a half-one metric:*

$$\text{ex}(\bar{\mathcal{M}}_n) \cap [\frac{1}{2}, 1]^{(n)} = \text{ex}(\mathcal{H}_n).$$

*Proof.* The companion paper shows  $\bar{\mathcal{M}}_n \cap [\frac{1}{2}, 1]^{(n)} = [\frac{1}{2}, 1]^{(n)}$ : any vector with all entries at least  $\frac{1}{2}$  satisfies every triangle inequality, since two sides sum to at least 1. An extreme point of  $\bar{\mathcal{M}}_n$  lying in a convex subset of  $\bar{\mathcal{M}}_n$  is extreme in that subset, so a vertex of  $\bar{\mathcal{M}}_n$  in the upper half is a vertex of the cube, that is, a point of  $\{\frac{1}{2}, 1\}^{(n)} = \mathcal{H}_n$ . The reverse containment is trivial.

The following direct argument avoids the cube identity and locates the failure. Let  $d$  be extreme with all values in  $[\frac{1}{2}, 1]$  and suppose some value  $v = d_E \in (\frac{1}{2}, 1)$ . The coordinate  $E$  then lies in no tight constraint of  $\bar{\mathcal{M}}_n$  at  $d$ : not in a bound, since  $0 < v < 1$ ; not as the long side of a degenerate triangle, since the two short sides sum to at least  $1 > v$ ; and not as a short side, since then the long side would be  $v + z \geq v + \frac{1}{2} > 1$ . Every constraint involving  $E$  being slack, the coordinate perturbation  $\pm \delta e_E$  preserves membership in  $\bar{\mathcal{M}}_n$  for small  $\delta > 0$ , contradicting extremality.  $\square$

So the vertex set of the upper half-cube of [5] is completely determined, and every vertex of  $\bar{\mathcal{M}}_n$  beyond the half-one family has some distance strictly below  $\frac{1}{2}$ —as the census of the next section will exhibit in quantity.

**Corollary 3.11** (why half and one). *Let  $V \subseteq (0, 1]$  satisfy  $2 \inf V \geq \sup V$ , so that every vector in  $V^{(n)}$  is a metric. Then*

$$\text{ex}(\bar{\mathcal{M}}_n) \cap V^{(n)} = \text{ex}(\mathcal{H}_n) \cap V^{(n)}.$$

*In particular the intersection is empty if  $1 \notin V$ , equals  $\{d \equiv 1\}$  if  $1 \in V$  but  $\frac{1}{2} \notin V$ , and  $\{\frac{1}{2}, 1\}$  is the only automatically feasible value set yielding anything more.*

*Proof.* Let  $d \in V^{(n)}$  be extreme and let  $M$  be its largest value. If  $M < 1$  then the scaling perturbation  $\pm \delta d$  preserves  $\bar{\mathcal{M}}_n$  for small  $\delta$ : bounds have slack, and the triangle constraints are homogeneous; so  $d$  is not extreme. Hence  $M = 1$ , so  $\sup V = 1$  and the hypothesis puts  $V \subseteq [\frac{1}{2}, 1]$ , and Theorem 3.10 applies. For the last clause: the value sets  $\{\frac{1}{k}, 1\}$  with  $k \geq 3$  are excluded by the hypothesis, and genuinely fail it, the vector with one unital pair and the rest  $\frac{1}{k}$  violating the triangle inequality already at  $n = 3$ .  $\square$

## 4 The vertices of $\bar{\mathcal{M}}_n$ for small $n$

A vertex of  $\bar{\mathcal{M}}_n$  is a rational point, being the unique solution of a rational linear system, and its denominator is bounded. One may therefore enumerate the vertices of denominator dividing  $L$  exhaustively: run over  $\{0, \frac{1}{L}, \dots, 1\}^m$  with the triangle inequalities used to prune, and certify each surviving point by computing the rank of its tight constraint system. Taking  $L$  through the denominators that occur settles  $|\text{ex}(\bar{\mathcal{M}}_n)|$ .

**Theorem 4.1.**  $|\text{ex}(\bar{\mathcal{M}}_4)| = 19$ ,  $|\text{ex}(\bar{\mathcal{M}}_5)| = 259$  and  $|\text{ex}(\bar{\mathcal{M}}_6)| = 27263$ . *The denominators occurring are  $\{1, 2\}$  for  $n \leq 5$  and  $\{1, 2, 3, 4\}$  for  $n = 6$ , with multiplicities*

| den     | 1   | 2     | 3    | 4    |
|---------|-----|-------|------|------|
| $n = 6$ | 203 | 15090 | 7290 | 4680 |

The four sweeps cross-check: at  $n = 6$  the grids  $L = 2, 3, 4$  return 15293, 7493 and 19973 vertices, which are the cumulative sums  $203 + 15090$ ,  $203 + 7290$  and  $203 + 15090 + 4680$  of the table. At  $n = 5$  the  $L = 4$  sweep returns the same 259 as  $L = 2$ , confirming that no vertex of denominator 3 or 4 is missed there, and at  $n = 4$  the  $L = 6$  sweep returns the same 19 as  $L = 2$ .

**Proposition 4.2.** *The positive-definite vertices of  $\bar{\mathcal{M}}_n$  of denominator dividing 2 are exactly the extreme half-one metrics.*

*Proof.* A vertex of denominator dividing 2 takes values in  $\{0, \frac{1}{2}, 1\}$ , and is positive definite exactly when the value 0 is unused, that is, when it lies in  $\{\frac{1}{2}, 1\}^{\binom{n}{2}} = \mathcal{H}_n$ .  $\square$

Accordingly the census returns 5, 168 and 12326 positive-definite denominator-2 vertices for  $n = 4, 5, 6$ , matching the enumeration of Remark 3.2 exactly and by an independent route.

The vertices that are not positive definite are accounted for by the positive-definite ones on fewer points, so the census reduces to a single sequence.

**Proposition 4.3.** *Write  $\pi_d$  for the partition of  $\mathbf{n}$  induced by  $d_{xy} = 0$  and  $\text{pd}(k)$  for the number of positive-definite vertices of  $\bar{\mathcal{M}}_k$ . Then  $d$  is a vertex of  $\bar{\mathcal{M}}_n$  if and only if the induced metric on the quotient  $\mathbf{n}/\pi_d$  is a positive-definite vertex there, and consequently*

$$|\text{ex}(\bar{\mathcal{M}}_n)| = \sum_{k=1}^n S(n, k) \text{pd}(k),$$

$S(n, k)$  the Stirling numbers of the second kind.

*Proof.* A perturbation  $\varepsilon$  at  $d$  satisfies  $|\varepsilon| \leq \min\{d, 1 - d\}$  by Lemma 2.1(i), so it vanishes on every pair inside a block of  $\pi_d$ . If  $x, x'$  lie in one block and  $y$  in another, the triangle on  $x, x', y$  is degenerate, so additivity gives  $\varepsilon_{xy} = \varepsilon_{xx'} + \varepsilon_{x'y} = \varepsilon_{x'y}$ : the perturbation is constant on blocks and descends. Perturbations of  $d$  and of the induced quotient metric therefore correspond, and extremality is preserved in both directions. Summing over the  $S(n, k)$  partitions with  $k$  blocks gives the count.  $\square$

So  $\text{pd}(1) = \text{pd}(2) = \text{pd}(3) = 1$ ,  $\text{pd}(4) = 5$ ,  $\text{pd}(5) = 168$ ,  $\text{pd}(6) = 24296$ , and Theorem 4.1 is recovered as  $1+7+6+5 = 19$ ,  $1+15+25+50+168 = 259$  and  $1+31+90+325+2520+24296 = 27263$ . This furnishes a check on the census independent of the sweeps: the 68 orbit representatives of the positive-definite vertices of  $\bar{\mathcal{M}}_6$  tabulated in [6, §5.1] expand to 24296 points under  $S_6$ , and the identity above then returns 27263, so no orbit is missing from that table and none is redundant. The same expansion splits the denominator-2 column of Theorem 4.1 as  $15090 = 12325 + 65 \cdot 4 + 15 \cdot 167$ , the last two terms being the denominator-2 vertices carried up from  $\bar{\mathcal{M}}_4$  and  $\bar{\mathcal{M}}_5$ , and identifies the denominator-1 column as the partition metrics,  $203 = B_6$ .

Comparing the two counts answers reading (ii) of the introduction:

| $n$ | $ \text{ex}(\bar{\mathcal{M}}_n) $ | $ \text{ex}(\mathcal{H}_n) $ | share |
|-----|------------------------------------|------------------------------|-------|
| 4   | 19                                 | 5                            | 0.263 |
| 5   | 259                                | 168                          | 0.649 |
| 6   | 27263                              | 12326                        | 0.452 |

What is known about the growth of  $|\text{ex}(\bar{\mathcal{M}}_n)|$  brackets it between two exponents.

**Proposition 4.4.**

$$(1 - o(1)) \binom{n}{2} \leq \log_2 |\text{ex}(\bar{\mathcal{M}}_n)| \leq \left(\frac{1}{2} + o(1)\right) \binom{n}{2} \log_2 n.$$



*Proof.* The lower bound is Theorem 3.1. For the upper bound,  $\bar{\mathcal{M}}_n$  is an  $m$ -polytope,  $m = \binom{n}{2}$ , with at most  $3\binom{n}{3} + \binom{n}{2} = (n-1)\binom{n}{2}$  facets: the triangle inequalities and the caps  $d_E \leq 1$ , the bounds  $d_E \geq 0$  being implied by sums of triangle inequalities for  $n \geq 3$ . By the upper bound theorem of McMullen [7], an  $m$ -polytope with  $N$  facets has at most  $\binom{N - \lceil m/2 \rceil}{\lfloor m/2 \rfloor} + \binom{N - \lfloor m/2 \rfloor - 1}{\lceil m/2 \rceil - 1} \leq 2\binom{N}{\lfloor m/2 \rfloor}$  vertices, and  $\binom{N}{k} \leq (eN/k)^k$  with  $N = (n-1)m$  and  $k = \lfloor m/2 \rfloor$  gives  $\log_2 |\text{ex}(\bar{\mathcal{M}}_n)| \leq \frac{m}{2} \log_2 n (1 + o(1))$ .  $\square$

At  $n = 6$  the asymptotic expressions evaluate to 15 and 19.4, while  $\log_2 27263 \approx 14.73$  sits below both: the  $o(1)$  corrections still dominate at this size, and the count has not yet chosen between the exponents.

*Question 4.5.* Does  $|\text{ex}(\mathcal{H}_n)|/|\text{ex}(\bar{\mathcal{M}}_n)|$  tend to a limit, and if so to what? The three available values are not monotone. Since  $|\text{ex}(\mathcal{H}_n)| = (1 - o(1))2^{\binom{n}{2}}$  by Theorem 3.1, the share tends to a positive limit or to zero according as  $\log_2 |\text{ex}(\bar{\mathcal{M}}_n)| - \binom{n}{2}$  stays bounded or tends to infinity, and by Proposition 4.4 either remains possible: the question is whether the logarithmic factor in the upper bound is real.

Question 4.5 is the point at which readings (ii) and (iii) may part company: the half-one metrics could carry asymptotically all of the exterior angle while forming a vanishing fraction of the vertices, if the remaining vertices proliferate with small angles. Nothing in our data excludes this, and nothing supports it.

## 5 Which denominators occur

Beyond  $n = 6$  the census is out of reach, but the denominators may be sampled. Solving  $10^3$  random linear objectives over  $\bar{\mathcal{M}}_n$  and rationalizing the optima exactly gives denominator sets  $\{1, 2\}$  for  $n \leq 5$ ,  $\{1, 2, 3, 4\}$  at  $n = 6$ ,  $\{1, \dots, 6\}$  at  $n = 7$  and  $\{1, \dots, 7\}$  at  $n = 8$ , in agreement with Theorem 4.1 where both apply.

A second question concerns not which denominators exist but which are needed. Write  $\bar{\mathcal{M}}_n^{\geq 1/2} = \bar{\mathcal{M}}_n \cap [\frac{1}{2}, 1]^m$  for the upper half of the body; by [5] this is exactly the cube  $[\frac{1}{2}, 1]^m$ , every point of which is a convex combination of at most  $m + 1$  half-one metrics, and every half-one metric is in turn a convex combination of vertices of the minimal face containing it. Carrying out the second step by repeated bisection of that face—at the current point take a direction in the lineality space of its smallest face, walk to both boundaries, split, recurse—produces a decomposition whose leaves are vertices of  $\bar{\mathcal{M}}_n$ .

Each bisection multiplies the denominator by at most 3, since the constraints of the face have at most three terms with coefficients  $\pm 1$ , so a face of dimension  $N$  yields denominators of the form  $2^i 3^j$  with  $i + j \leq N + 1$ . What is observed is far smaller.

*Conjecture 5.1.* Every  $d \in \bar{\mathcal{M}}_n^{\geq 1/2}$  is a convex combination of extreme points of  $\bar{\mathcal{M}}_n$  whose denominators divide 12, uniformly in  $n$ .

Running the bisection and recording every leaf gives, exhaustively for  $n \leq 5$  and by sampling for  $6 \leq n \leq 10$  with faces of dimension up to 13, only the denominators 1, 2, 3, 4, 6: never 8, 9 or 12, and never a denominator with a prime factor other than 2 or 3. The obstruction to a proof is compounding: nothing in the argument forbids a coefficient 3 at a second bisection after a 2 or a 3 at the first, and yet it does not occur. A slack-parity invariant would settle it.

One tempting sharpening is false: denominator 3 alone does not suffice. Two computations say so, at different strengths.

Globally, take all points of  $[0, 1]^m$  with coordinates in  $\{0, \frac{1}{3}, \frac{2}{3}, 1\}$  that are metrics—at  $n = 5$  there are 28818 of them, a set containing every vertex of  $\bar{\mathcal{M}}_5$  of denominator dividing 3 and much

besides. Even so, 255 of the 856 non-extreme half-one metrics at  $n = 5$  lie outside their convex hull. Restricting the generators to the vertices themselves, which at  $n = 5$  are just the 52 partition metrics since denominator 3 does not occur there, raises the count to 375.

Locally, one may ask whether the decomposition can be carried out inside the minimal face of  $d$ , moving only along its bipartite components. Pushing each such component to  $\pm\frac{1}{6}$  or  $\pm\frac{1}{2}$  produces candidate vertices with values in  $\{0, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, 1\}$ ; 180 of the 856 at  $n = 5$ , and 593 of a sample of 2527 at  $n = 6$ , are not convex combinations of the candidates their own face provides.

Either way the denominator-2 vertices, those with values in  $\{0, \frac{1}{2}, 1\}$ , are genuinely needed.

## 6 $q$ -level points and the scarcity of other denominators

Half-one metrics are automatically feasible; points of other denominators are not, and the following measures how badly.

Call the points of  $[0, 1]^m$  with coordinates in  $\{0, \frac{1}{q-1}, \dots, 1\}$  the  $q$ -level points, so that a  $q$ -level point has denominator dividing  $q - 1$ .

**Proposition 6.1.** *A triple of levels drawn uniformly at random satisfies the triangle inequality with probability*

$$P_q = \frac{1}{2} + \frac{1}{2q^2}.$$

*Proof.* Scale the levels to  $\{0, 1, \dots, q - 1\}$  and fix  $a_3 = k$ . The ordered pairs  $(a_1, a_2)$  of levels with  $a_1 + a_2 < k$  number  $\binom{k+1}{2}$ . Summing over  $k = 0, \dots, q - 1$  gives  $\binom{q+1}{3}$  infeasible ordered triples for the inequality  $a_3 \leq a_1 + a_2$ ; a triple failing one of the three inequalities passes the other two, so the total infeasible count is  $3\binom{q+1}{3} = \frac{q(q+1)(q-1)}{2}$  out of  $q^3$ , and  $P_q = 1 - \frac{(q+1)(q-1)}{2q^2} = \frac{1}{2} + \frac{1}{2q^2}$ .  $\square$

So  $P_q \rightarrow \frac{1}{2} = \text{Vol}(\bar{\mathcal{M}}_3)$ , and from above: the lattice slightly overestimates the volume at every finite level. The corresponding formula at [6, (5.2.1)] carries the opposite sign; the counting argument there is the one reproduced above, and it requires 0 to be an admissible level, which fixes the convention and the direction of approach. Combining with Spencer's bound on families of independent triples [9]:

**Corollary 6.2.**  $\text{Vol}(\bar{\mathcal{M}}_n) \leq 2\left(\frac{1}{\sqrt[3]{2}}\right)^{\binom{n}{2}} \rightarrow 0$  as  $n \rightarrow \infty$ .

Monte Carlo gives  $\text{Vol}(\bar{\mathcal{M}}_n) = 0.5007, 0.1418, 0.0225, 0.00196, 0.00009$  for  $n = 3, \dots, 7$ , against bounds 1, 0.5, 0.198, 0.0625, 0.0156.

Feasibility and degeneracy pull against each other, and the following is the extreme case.

**Proposition 6.3.** *Let  $d \in \mathbb{R}^m$  be strictly positive and suppose every triple of points carries a degenerate triangle. Then  $d$  is a metric.*

*Proof.* If  $d_{ik} = d_{ij} + d_{jk}$  then  $d_{ik}$  is the largest of the three and all three triangle inequalities on that triple hold.  $\square$

By Proposition 2.2 each triple carries at most one degenerate ordered triangle, so  $\binom{n}{3}$  is the maximum possible count and Proposition 6.3 concerns exactly the maximizers.

For 3-level points the tension is visible experimentally: Figure 2 plots degenerate against feasible triangle counts for random 3-level points. As  $n$  grows the distribution becomes approximately Gaussian, consistent with the triangles becoming asymptotically independent, and feasibility is lost as degeneracy grows. Figure 1 shows the progression toward normality in the feasible-triangle count alone.

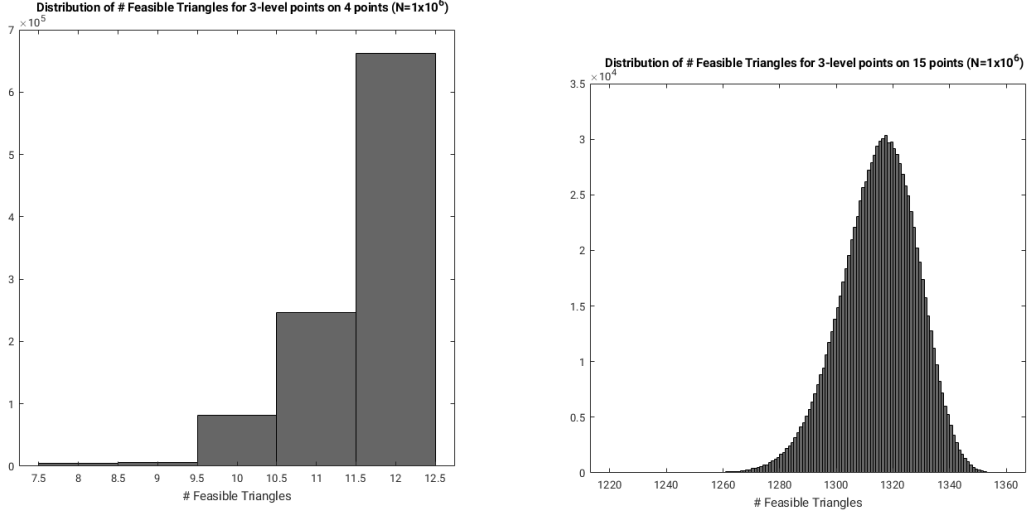


Figure 1: Distribution of the number of feasible triangles among uniformly drawn 3-level points,  $n = 4$  (left) and  $n = 15$  (right).

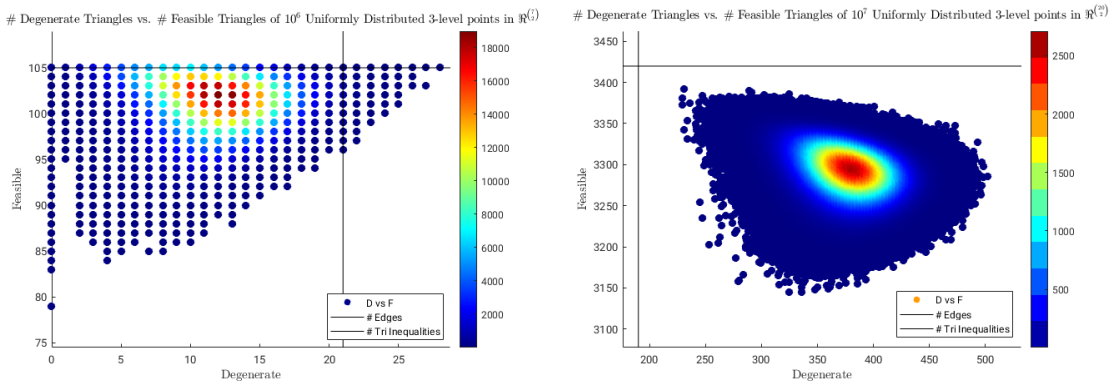


Figure 2: Degenerate versus feasible triangle counts for random 3-level points,  $n = 7$  (left) and  $n = 20$  (right).

The bowtie family evades the tension by anchoring the degeneracies on a single unital stem. Let

$$\mathcal{B}_{n,1}^3 = \{d \in I_n^3 : \exists! I \in \mathcal{E} \text{ with } d_I = 1, \text{ and every triangle containing } I \text{ is degenerate}\}.$$

Every member is a metric—the infeasible pattern  $(\frac{1}{3}, \frac{1}{3}, 1)$  is excluded—and the degenerate anchor lets the  $(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$  degeneracies propagate.

## 7 Exterior angles

The probability that a vertex  $d^*$  maximizes a uniformly random linear objective is proportional to the volume of its dual cone met with the unit sphere: the exterior solid polyhedral angle at  $d^*$ . Sampling random objectives therefore samples vertices by angle.

| $n$ | partition | half-one | den $\leq 2$ | other |
|-----|-----------|----------|--------------|-------|
| 5   | 0.978     | 0.000    | 1.000        | 0.000 |
| 8   | 0.870     | 0.007    | 0.987        | 0.013 |
| 10  | 0.657     | 0.003    | 0.923        | 0.077 |
| 14  | 0.310     | 0.040    | 0.640        | 0.360 |
| 18  | 0.093     | 0.107    | 0.467        | 0.533 |
| 20  | 0.050     | 0.275    | 0.625        | 0.375 |
| 24  | 0.000     | 0.530    | 0.830        | 0.170 |
| 32  | 0.000     | 0.967    | 1.000        | 0.000 |
| 40  | 0.000     | 1.000    | 1.000        | 0.000 |

The half-one share is flat near zero through  $n = 12$ , turns near  $n = 14$ , crosses the partition share near  $n = 18$ , and reaches 1 by  $n = 40$ . An experiment confined to  $5 \leq n \leq 10$  sees only the flat initial segment, and would report no trend.

*Conjecture 7.1* (Gauss–Bonnet). The sum of the exterior solid polyhedral angles over the extreme half-one metrics on  $n$  points converges, as  $n \rightarrow \infty$ , to the volume of the  $\binom{n}{2}$ -dimensional unit sphere.

The name records the analogy: the total exterior angle of a polytope is its whole sphere of directions, and the conjecture asserts that the half-one vertices absorb all of that curvature in the limit. The heuristic is that an extreme half-one metric lies on many more supporting hyperplanes than extremality requires, having many degenerate triangles, and is therefore a sharp corner with a large dual cone.

Three abundance statements should be kept apart. Theorem 3.1 says half-one metrics are almost all extreme; the table above says they win the angle competition from about  $n = 20$ ; Conjecture 7.1 says they take all of it in the limit; and Question 4.5 asks about their share of the vertices, where the evidence is equivocal. The first is proved, the second observed, the third open, and the fourth may have a different answer from the third.

A reduction worth recording: the total angle carried by non-half-one vertices is at most the number of such vertices times the maximal angle at one of them, so any subexponential bound on the maximal exterior angle at a non-half-one vertex, combined with a bound on  $|\text{ex}(\bar{\mathcal{M}}_n)|$ , would bear on Conjecture 7.1. Proving it appears to require estimating the hypergeometric series arising from the angular geometry of the  $e_{[i,j,k]}$ -system of supporting hyperplanes.

## 8 Questions

*Question 8.1.* Prove the upper bound matching Theorem 3.4: that  $1 - |\text{ex}(\mathcal{H}_n)|/2^{\binom{n}{2}} = O(n^2 2^{-n})$ , establishing Conjecture 3.5. What is required is a union bound over bipartite components of  $\Gamma_d$  with at least two nodes; the classification through  $n = 8$  shows every such mode decaying at least at the rate  $4^{-n}$  against the twin rate  $2^{-n}$ , but the number of shapes grows and no uniform bound is proved.

Question 4.5 above asks for the growth of  $|\text{ex}(\bar{\mathcal{M}}_n)|$ , and Conjecture 5.1 for a denominator bound uniform in  $n$ . Two further questions:

*Question 8.2.* Is  $|\text{ex}(\bar{\mathcal{M}}_7)|$  computable by the method of Section 4? The denominator-2 sweep at  $n = 7$  is within reach and would extend Proposition 4.2 to 1309868; the full census requires denominators to 6 and a better enumerator than exhaustive search over grids.

*Question 8.3.* Theorem 3.10 determines the vertices of  $\bar{\mathcal{M}}_n$  in the closed upper half-cube. The census shows the vertices below it organized by denominator; is there a second value-set theorem there—a description of  $\text{ex}(\bar{\mathcal{M}}_n) \cap [\frac{1}{3}, 1]^{\binom{n}{2}}$ , say, where degenerate triangles of patterns other than  $\frac{1}{2} + \frac{1}{2} = 1$  first become available?

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